

Algorithmic Explanations and Institutional Legitimacy in AI-Enabled Governance

A 2 x 3 Between-Subjects Experiment

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2026-10-05

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1 1. Introduction

Governments increasingly turn to algorithmic systems for consequential decisions—welfare eligibility, parking fines, law enforcement. These systems promise consistency and efficiency, and agencies often assume that explaining a decision will keep citizens on board. But explanations do not reliably produce legitimacy: people can read an explanation and still judge the decision unfair or even dehumanizing. This study compares three ways of explaining an algorithmic decision—stating only the outcome, walking through the process, or adding empathic framing—and asks how each shapes perceived institutional legitimacy, and whether the effect depends on how much is at stake. In a 2 x 3 between-subjects experiment ($N = 420$), we measure perceived institutional legitimacy, institutional trust, and policy acceptance.

2 2. Method

We ran a 2 (decision stakes: low vs. high) x 3 (explanation type: outcome, process, empathic) between-subjects experiment, with 70 participants per cell ($N = 420$). The low-stakes vignette described an automated parking fine; the high-stakes vignette described the denial of public housing assistance. Outcome explanations reported only the decision; process explanations described the rules behind it; empathic explanations added a note of concern and mentioned a human review channel. Participants rated perceived institutional legitimacy (4 items), institutional trust (3 items), and policy acceptance (3 items) on 5-point scales, and also completed control measures of government trust, AI attitude, AI experience, and policy attention. We report internal consistency (Cronbach’s alpha), descriptive statistics, and 2 x 3 ANOVAs with partial eta-squared, followed by simple-effects tests and Tukey HSD where interactions were significant.

3 3. Results

All three outcome scales showed good internal consistency (Table 1).

Table 1: Internal consistency of the outcome scales

Scale	Items	Cronbach’s alpha
Perceived institutional legitimacy (PIL)	4	0.92
Institutional trust (IT)	3	0.88
Policy acceptance (PA)	3	0.91

Descriptive statistics by condition are reported in Table 2.

Table 2: Means (standard deviations) by condition

Decision stakes	Explanation type	n	Legitimacy	Trust	Acceptance
Low Stakes	Outcome	70	4.06 (0.73)	4.18 (0.70)	4.06 (0.79)
Low Stakes	Process	70	4.05 (0.53)	4.21 (0.42)	4.02 (0.62)
Low Stakes	Empathic	70	3.98 (0.70)	4.20 (0.56)	4.01 (0.77)
High Stakes	Outcome	70	3.11 (1.08)	3.55 (1.16)	3.15 (1.24)
High Stakes	Process	70	3.25 (1.14)	3.70 (0.98)	3.25 (1.16)
High Stakes	Empathic	70	2.48 (1.01)	3.02 (1.07)	2.38 (1.13)

Decision stakes mattered a great deal. High-stakes decisions produced clearly lower perceived legitimacy ($F(1, 414) = 154.55$, $p < .001$, $\eta^2 = .27$), trust ($F = 85.67$, $p < .001$, $\eta^2 = .17$), and

acceptance ($F = 133.84$, $p < .001$, $\eta^2 = .24$). Explanation type also mattered, but the effects were modest (all $p < .01$, $\eta^2 = .04$).

The two factors interacted for every outcome (all $p < .01$, $\eta^2 = .03$; Table 3, Figure 1). Explanation type made no difference under low stakes (all $p > .72$), but a clear difference under high stakes (all $p < .001$). Under high stakes, empathic explanations produced the lowest legitimacy ($M = 2.48$), below outcome ($M = 3.11$) and process ($M = 3.25$); Tukey tests confirmed empathic differed from both ($ps < .01$), while outcome and process did not ($p > .70$).

Table 3: 2 x 3 ANOVA results with partial eta-squared

Outcome	Effect	F	p	partial η^2
Perceived institutional legitimacy	Stakes	154.55	$< .001$	0.272
Perceived institutional legitimacy	Explanation	8.98	$< .001$	0.042
Perceived institutional legitimacy	Stakes x Explanation	5.91	.003	0.028
Institutional trust	Stakes	85.67	$< .001$	0.171
Institutional trust	Explanation	5.95	.003	0.028
Institutional trust	Stakes x Explanation	6.12	.002	0.029
Policy acceptance	Stakes	133.84	$< .001$	0.244
Policy acceptance	Explanation	8.8	$< .001$	0.041
Policy acceptance	Stakes x Explanation	7.9	$< .001$	0.037

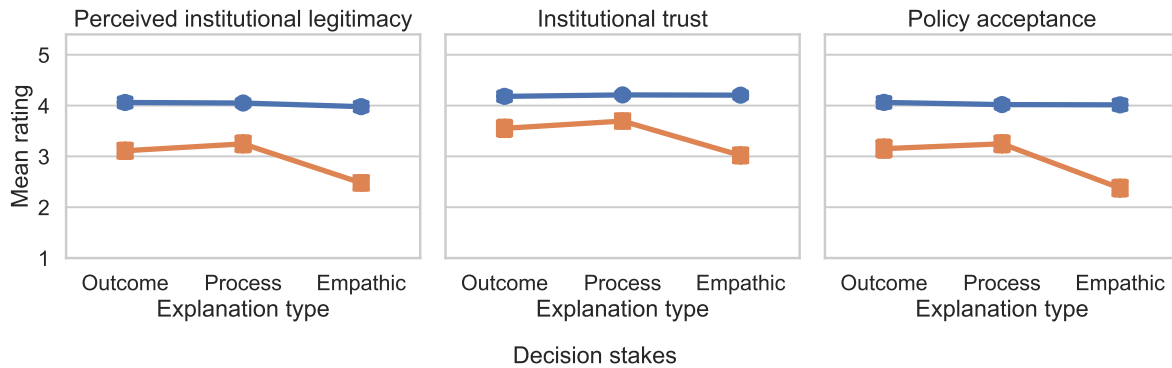


Figure 1: Mean ratings by explanation type and decision stakes (error bars = ± 1 SE)

4 4. Discussion

The clearest finding is that empathic explanation backfired exactly where it should have helped most. In high-stakes settings, the version that added human concern and a review channel

produced the lowest legitimacy, trust, and acceptance. One reading is that when a decision carries real consequences, people want to see the rules—not reassurance—and a sympathetic tone can read as insincere, or as a way of softening the blow rather than justifying it. For low-stakes decisions, the explanation hardly mattered at all.

That has practical implications. For routine decisions, agencies can spend less effort on explanation. For high-stakes decisions, process explanations look safest, while empathic framing should be used cautiously, since it may undermine rather than rebuild legitimacy.

The study has limits worth keeping in mind. We did not measure the mechanisms—procedural fairness, dehumanization—or the moderators the broader project proposes, so the psychological account is speculative. The sample is a convenience survey, and each stakes level used a single vignette. Follow-up work should measure those mechanisms directly and test whether the backfire effect holds for real, consequential decisions.